PCRQ 14

Graded

Student

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Total Points

5 / 6 pts

Question 1

(no title) — **0** / 1 pt

✔ **+ 0 pts** Incorrect

Question 2

(no title) — **1** / 1 pt

✔ **+ 1 pt** Correct

Question 3

(no title) — **1** / 1 pt

✔ **+ 1 pt** Correct

Question 4

(no title) — **1** / 1 pt

✔ **+ 1 pt** Correct

Question 5

(no title) — **1** / 1 pt

✔ **+ 1 pt** Correct

Question 6

(no title) — **1** / 1 pt

✔ **+ 1 pt** good faith answer

For a sequence that converges to 2, what convergence tells us is that any epsilon ball around 2, here any interval around 2, should contain all except possibly finitely many the terms of the sequence. So an interval of the form (2-epsilon, 2+epsilon) must contain all but finitely many terms of the sequence. This means that its complement must be finite, and any subset of the complement must therefore also be finite. Now when it comes to A_5 we don't know whether it is finite or infinite because the sequence may be approaching 2 from the right---- and then the interval (2-epsilon, 2+epsilon) would still have all but finitely many the terms of the sequence a_n but they would all be in (2, 2+epsilon). So it may be the case that A_5 is finite or that it is infinite depending on whether the sequence approaches 2 from the left, from the right, or oscillates around 2 approaching it from both sides. For the squeeze theorem proof we are able to capture the limit of b_n because it is upper and lower bounded by two sequences that have the same limit: so we have to get to n big enough so that both a_n and c_n are in a strip of radius epsilon around the limit, and then say that b_n will also have to be in that strip since it should satisfy a_n \leq b_n \leq c_n.

Q1
1 Point

Suppose (a_n) is a sequence of real numbers with

$$\lim_{n\to\infty} a_n = 2.$$

Consider the sets

$$A_1 = \{n \in \mathbb{N} : a_n < 2.001\}, \qquad A_2 = \{n \in \mathbb{N} : a_n > 2.003\},$$
$$A_3 = \{n \in \mathbb{N} : a_n < 1.98\}, \qquad A_4 = \{n \in \mathbb{N} : 1.99997 < a_n < 2.0001\},$$

and

$$A_5 = \{n \in \mathbb{N} : a_n \leq 2\}.$$

Which of the following correctly describes what MUST be true?

- (●) A_1, A_4, A_5 have finite complements, while A_2, A_3 are finite.
- ( ) A_1, A_4 have finite complements, A_2, A_3 are finite, and neither conclusion is necessarily true for A_5.
- ( ) A_1, A_4 are finite, while A_2, A_3 have finite complements, and A_5 is finite.
- ( ) All five sets have finite complements.

Q2
1 Point

Every bounded sequence of real numbers converges.

- ( ) True
- (●) False

Q3
1 Point

Every convergent sequence is bounded.

- (●) True
- ( ) False

Q4
1 Point

If (a_n) is bounded and (b_n) converges, then $(a_n b_n)$ must converge.

- ○ True
- ◉ False

Q5
1 Point

Suppose $a_n \to a$, $b_n \to b$, and

$$a_n < b_n$$

for all sufficiently large n. What can we conclude?

- ○ $a < b$
- ◉ $a \leq b$
- ○ $a = b$
- ○ Nothing can be concluded about a and b.

Q6
1 Point

Please write briefly (in two or three sentences) what single point of the notes of Lecture 15 you found to be the most difficult or confusing. If you did not find any part of it difficult or confusing, please say what part(s) you found most interesting.

Can you help explain more on Q1, how to know when a set has finite or infinite complement there?

For the proof of the Squeeze Theorem, why is it necessary to let N_0=max{N_1, N_2}, also why maximum.